

A15536W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics, and Economics
Honour School of Economics and Management
Honour School of History and Economics

MICROECONOMIC ANALYSIS

TRINITY TERM 2024

Tuesday 04 June, 09:30–12:30

Please start the answer to each question on a separate page.

There are 6 questions in this paper.

*Answer **four** questions.*

For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

1. (a) Determine the solutions of the system below, depending on the value of $\alpha \in \mathbb{R}$.

$$\begin{cases} x + 2y - 3z &= 4 \\ 4x + y + 2z &= 6 \\ x + 2y + (\alpha^2 - 19)z &= \alpha \end{cases}$$

[25%]

- (b) Prove formally, by contradiction, that the following statement is true:

If $v_1, v_2, v_3 \in \mathbb{R}^3$ satisfy: (i) $v_1 \neq (0, 0, 0)$, (ii) $v_2 \notin \text{span}(v_1)$, (iii) $v_3 \notin \text{span}(v_1, v_2)$, then v_1, v_2, v_3 are linearly independent.

Under these conditions, would you be able to conclude that these vectors form a basis of $\mathbb{R}^3$? [25%]

- (c) Consider the function $f(x, y) = \frac{x^3 + y}{1 + 2xy}$ whenever $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Show that, for every line, if you analyse the limit of a sequence of points in that line approaching $(0, 0)$, the limit is always zero. Is the function continuous at the origin? [25%]

- (d) Considering the solutions to the system below, is it possible to define (x, y) as a function of z in a neighborhood of the point $(1, 2, 2)$? If so, what are the derivatives of $x(z)$ and $y(z)$ at $z = 2$?

$$\begin{cases} x^3y - z &= 0 \\ x + y^2 + z^3 &= 13 \end{cases}$$

[25%]

2. Suppose that $X = \mathbb{R}$ represents all possible locations in a (very narrow) town. There are two schools in that town, with locations $s_1 < s_2$. Families are looking for a place to live. They all have the following preferences:

$$x \succsim y \Leftrightarrow \min\{|x - s_1|, 3|x - s_2|\} \leq \min\{|y - s_1|, 3|y - s_2|\}.$$

- (a) Show that these preferences satisfy completeness and transitivity. Give a simple interpretation of the preferences using the following two concepts: *commuting time* and *quality of the school*. Which is the best school in that interpretation? [25%]
- (b) Do these preferences admit a utility representation? Are these preferences single-peaked? [25%]

Suppose now that you have access to the following type of choice data: you have observed choices from some menus that are closed intervals $A = [l, r]$, with $l < r$. Assume in all cases that families choose by maximising the preferences presented.

- (c) Let A_1, A_2 be such that $l_1 < l_2 < r_2 < r_1$. Is it necessarily the case that choices in A_1, A_2 are the same? What if we know that the choice in menu A_1 belongs to menu A_2 ? [25%]
- (d) By considering all possible orders over $s_1 < s_2$ and $l < r$, explain the choices for any menu $A = [l, r]$. You can just assume that the orders are strict, i.e. l and r are different from s_1, s_2 . You can also assume that, if there is any case with two optimal locations, any of them is chosen. [25%]

3. Consider the space of monetary payouts $[1, e]$ (that is, the minimum payout is 1, and the maximum payout is number e ; recall that $\log 1 = 0$ and $\log e = 1$). An individual has expected utility preferences, with monetary utility $u(x) = \frac{-1}{x^2}$. The wealth of the individual is random. The density function defining the wealth lottery L of this individual is $f(x) = ax$.
- (a) Show that the constant a must be equal to $\frac{2}{e^2-1}$. [20%]
- (b) Compute the expected value, the expected utility and the certainty equivalent of the lottery. Recall that the integral of $\frac{1}{x}$ is $\log x$. [30%]
- (c) Using lottery L and payouts 1 and e , explain the property of continuity. What is the probability for which this property holds in the case of lottery L ? [20%]
- (d) The individual is asked to express a preference over three different options. Option 1 is to keep L . Option 2 involves flipping a fair coin: if the result is heads, the individual keeps L ; if the result is tails, the individual will receive the certainty equivalent of lottery L with certainty. Option 3 involves flipping a fair coin: if the result is heads, the individual receives the expected value of L ; if the result is tails, the individual receives wealth e with certainty. Without computations, using simple arguments based on basic properties of expected utility, discuss the preferences of the individual over these three options. [30%]

4. Consider the following function:

$$f(x, y) = x^2 + y^2 - axy + x + 2y$$

where $a \in \mathbb{R}$ is a parameter.

- (a) What does it mean for a function to be convex? For what values of a is the function $f(x, y)$ strictly convex? [20%]
- (b) Find global minimum of $f(x, a)$ for each value of parameter a (or establish that it does not exist). [20%]

For the rest of the question assume that $a = 0$. Consider the problem of finding the minimum of $f(x, y)$ subject to the following constraints:

$$x^2 + y^2 \leq r, \quad x \leq 0, \quad y \leq 0$$

where $r > 0$ is a parameter.

- (c) Plot the feasible region. [5%]
- (d) Argue why non-positivity constraints are not binding. [5%]
- (e) What does it mean that an optimization problem is convex? Is optimizing function $f(x, y)$ subject to these constraints a convex problem? [10%]
- (f) Write down the Kuhn-Tucker first order conditions for this problem. [10%]
- (g) Find the solution of the optimization problem for each value of r . Verify the constraint qualification. Argue why your solutions is indeed the minimum. [30%]

5. A risk-neutral fishing boat owner hires a risk-averse fisherman to operate his boat. The total catch x may be worth either 60, 100 or 150. The probability of each outcome depends on how hard the fisherman works. There are two possible levels of effort: e_1 (high) and e_0 (low). The probability of each realisation of catch conditional on effort is given in the following table:

effort/catch	60	100	150
e_0	1/3	1/2	1/6
e_1	1/6	1/2	1/3

The fisherman's utility from receiving a wage $w \geq 0$ and exerting effort e is $u = \sqrt{w} - c(e)$, where $c(e_0) = 0$ and $c(e_1) = c$. The fisherman's reservation utility equals $\bar{u}$. The boat owner's profit is $\pi = x - w$.

- (a) Suppose that both the effort and the catch are observable. For what values of $c > 0$ and $\bar{u} > 0$ does the boat owner want to implement high effort? [30%]

Now assume that effort is unobservable. Moreover, assume that $\bar{u} > 3c$.

- (b) Suppose that the boat owner wants to implement effort e_0 . What contract would he choose? [10%]
- (c) Suppose that the boat owner wants to implement effort e_1 . Write down the fisherman's participation and incentive compatibility constraints and set up boat owner's optimization problem. Derive the optimal contract implementing e_1 . [40%]
- (d) Suppose that $c = 1$ and $\bar{u} = 6$. What contracts would the principal offer under observable and unobservable efforts? [20%]

6. Pandora faces 10,000 boxes she could open. Suppose that the distribution of prizes in box n is given by a probability distribution $F_n = 1 - e^{-u/n}$, where e is the base of natural logarithm, i.e. $\log e = 1$, $u \geq 0$ is a prize value and $n = 1, \dots, 10000$. Each box is clearly labelled as “box n ”, so Pandora knows the distribution associated with each box. If Pandora decides not to open any of the boxes her utility is $u_0 = 0$. Suppose that Pandora can open box n at a cost s_n . Pandora may open boxes in any order she wants, and can stop at any moment and take the prize from the best opened box, or may decide not to open any boxes and take the outside option.

- (a) Find the expected value of prize in box n . How does it depend on n ? [10%]
- (b) Denote by z_n the value for which Pandora is indifferent between accepting it immediately and opening box n only. Derive z_n . [30%]
- (c) Formulate the optimal search rule. [10%]
- (d) Suppose that $s_n = an$, $a > 0$. Show that if $a < 1$ Pandora wants to open box 10,000 first. What is the optimal search rule in this case? What Pandora will do if (i) $a > 1$, (ii) $a = 1$? [20%]
- (e) Suppose that $s_n = \frac{1}{e}n$, and suppose that the *actual* prize in box n equals to $u_n = \frac{10000}{n}$. Which box Pandora will end up choosing? [30%]

1. (a) By Gaussian elimination, the first row is $(1, 2, -3, 4)$, the second row becomes $(0, -7, 14, -10)$, and the third row becomes $(0, 0, \alpha^2 - 16, \alpha - 4)$. Since $\alpha^2 - 16 = (\alpha - 4)(\alpha + 4)$, we have three cases:
 - $\alpha = -4$: the system is indeterminate and no solutions
 - $\alpha = 4$: the system has infinite solutions. We can express all of them in terms of z by solving for y in the second row, and then for x in the first row.
 - Any other value: the system has a unique solution. We can solve for $z = \frac{1}{\alpha+4}$ in the last equation, and then use the former expressions of x and y .
- (b) If they were not independent, there would be a_1, a_2, a_3 , with at least one non-zero, such that $a_1v_1 + a_2v_2 + a_3v_3 = 0$. Suppose that a_3 is non-zero, then we can write v_3 as a combination of v_1 and v_2 , contradicting (iii). If $a_3 = 0$ and a_2 non-zero, then we can write v_2 as a combination of v_1 , contradicting (ii). Otherwise, a_1 is the unique non-zero element, and $a_1v_1 = 0$ implies $v_1 = (0, 0, 0)$, contradicting (i). The result is true. Three independent vectors of $\mathbb{R}^3$ form a basis.
- (c) Consider $f(x, y) = \frac{x^3+y}{1+2xy}$. If $x = 0$, the function becomes y , that converges to 0 when y approaches 0. Otherwise, let $x \neq 0$ and write $y = kx$. It follows that $f(x, y) = \frac{x^3+y}{1+2xy} = \frac{x^3+kx}{1+2kx^2}$ that approaches to zero when x approaches zero. However, this is not enough to see that the function is continuous, and students must argue for every possible sequence: for any sequence $\{(x_n, y_n)\} \rightarrow (0, 0)$ we have $f(x_n, y_n) \rightarrow 0$, so $f(x, y)$ is continuous at $(0, 0)$.
- (d) First, it is clear that $(1, 2, 2)$ solves the equations. Second, the two functions involved are C^1 . Then, we just need to analyse the Jacobian at the point at hand. The first row of the Jacobian is $(3x^2y, x^3)$ and the second row is $(1, 2y)$. In the relevant point, these are $(6, 1)$ and $(1, 4)$. The determinant of this Jacobian is $23 \neq 0$, making the Jacobian invertible. The answer is then positive. The derivatives with respect to z are $(-1, 3z^2)$, i.e., $(-1, 12)$. The answer can be immediately obtained from the inverse of the Jacobian times $(-1, 12)$.

2. (a) For every x, y , $\min\{|x - s_1|, 3|x - s_2|\} \leq \min\{|y - s_1|, 3|y - s_2|\}$ or $\min\{|y - s_1|, 3|y - s_2|\} \leq \min\{|x - s_1|, 3|x - s_2|\}$, proving completeness. Similarly, for every x, y, z , if $\min\{|x - s_1|, 3|x - s_2|\} \leq \min\{|y - s_1|, 3|y - s_2|\}$ and $\min\{|y - s_1|, 3|y - s_2|\} \leq \min\{|z - s_1|, 3|z - s_2|\}$, it must be $\min\{|x - s_1|, 3|x - s_2|\} \leq \min\{|z - s_1|, 3|z - s_2|\}$ and transitivity holds. Preferences can be interpreted as a combination of commuting time and quality of the school. Notice that we can think of $|x - s_i|$ as the distance to school s_i (capturing commuting time, that wants to be minimised). Distances are weighted differently for both schools, as the second distance is multiplied by 3. This can reflect, e.g., that one school is better than the other and hence a trade-off with commuting time exists. The distance to school 2 matters more, indicating that the school is of lower quality.
- (b) The answer is yes, and a utility representation is trivial, just take the negative of the expression defining the preferences. It is not enough to say that preferences are rational, as this is only necessary. Preferences are not single-peaked. Notice that $s_1 > \frac{s_1 + s_2}{2} > s_2$ and both $s_1 \succ \frac{s_1 + s_2}{2}$ and $s_2 \succ \frac{s_1 + s_2}{2}$.
- (c) Choices can be different. Consider, e.g. $l_1 < l_2 < r_2 < r_1 < s_1$. Choice in A_i is r_i , and then they are different. The preferences are rational and satisfy property α . If the choice of A_1 belongs to A_2 , property α guarantees that the choice in A_2 must be $c(A_1)$.
- (d) We divide the analysis in several cases, depending on where l and r are.
- $l < r < s_1$: the choice is r
 - $l < s_1 < r < s_2$: the choice is s_1
 - $s_1 < l < r < s_2$: the choice is l if $|s_1 - l| \leq 3|s_2 - r|$ and r otherwise
 - $s_1 < l < s_2 < r$: the choice is s_2
 - $s_2 < l < r$: the choice is l

3. Consider the space of monetary payouts $[1, e]$ (that is, the minimum payout is 1, and the maximum payout is number e ; recall that $\log 1 = 0$ and $\log e = 1$). An individual has expected utility preferences, with monetary utility $u(x) = \frac{-1}{x^2}$. The wealth of the individual is random. The density function defining the wealth lottery L of this individual is $f(x) = ax$.

- (a) The total probability mass of the lottery must be equal to 1. That is

$$\int_1^e ax \, dx = \frac{ax^2}{2} \Big|_1^e = \frac{a}{2}(e^2 - 1) = 1$$

Hence, $a = \frac{2}{e^2 - 1}$.

- (b) The expected value is equal to

$$\int_1^e ax^2 \, dx = \frac{ax^3}{3} \Big|_1^e = \frac{a}{3}(e^3 - 1) = \frac{2(e^3 - 1)}{3(e^2 - 1)}$$

The expected utility of the lottery is

$$\int_1^e ax \frac{-1}{x^2} \, dx = \int_1^e a \frac{-1}{x} \, dx = -a \log x \Big|_1^e = -a = \frac{-2}{e^2 - 1}$$

The certainty equivalent of the lottery satisfies

$$\frac{-1}{CE^2} = -a \text{ and } CE = \sqrt{\frac{e^2 - 1}{2}}$$

- (c) Since e is the best payout and 1 is the worst, continuity requires that there exists a value $p \in (0, 1)$ such that the lottery in which the individual receives e with probability p and 1 with probability $1 - p$ is indifferent to L . Notice that the expected utility of the binary lottery is $p \frac{-1}{e^2} + (1 - p) \frac{-1}{1}$. The expected utility of lottery L is $-a$. Hence, it must be

$$\frac{2}{e^2 - 1} = p \frac{1}{e^2} + (1 - p)1$$

This leads to $p = \frac{1}{e^2 - 1}$.

- (d) Using independence, since L is indifferent to its certainty equivalent, options 1 and 2 are equally good. Now, notice that this individual is strictly risk averse (it can be seen that the Arrow-Pratt coefficient is equal to 2) and hence, the expected value is strictly better than the certainty equivalent. Payout e is also strictly better than the lottery and hence, independence guarantees that option 3 is strictly better than options 1 and 2.

4. (a) A function is convex on S if for all $\mathbf{x}, \mathbf{y} \in S$ $f(\alpha\mathbf{x} + (1-\alpha)\mathbf{y}) \leq \alpha f(\mathbf{x}) + (1-\alpha)f(\mathbf{y})$ with strict inequality for strict convexity. The Hessian of function f is $\begin{vmatrix} 2 & -a \\ -a & 2 \end{vmatrix} = 4 - a^2$. Hence, the function is strictly convex if $a \in (-2, 2)$.

- (b) Suppose that $a \in (-2, 2)$. Then the system of the first order conditions is

$$\begin{cases} 1 + 2x - ay = 0 \\ 2 - ax + 2y = 0 \end{cases} \rightarrow x^* = -\frac{2+2a}{4-a^2}, y^* = -\frac{4+a}{4-a^2},$$

which gives $f(x^*, y^*) = -\frac{5+2a}{4-a^2}$.

If $a \notin (-2, 2)$, then the function does not have global minimum: this can be verified by either taking $x = y = t$ or $x = -y = t$ and letting $t \rightarrow \pm\infty$.

- (c) It is a quarter of a circle with centre at $(0, 0)$ and radius $\sqrt{r}$, located in negative quadrant.
- (d) A possible argument would be that both $f_x(0, y) > 0$ and $f_y((x, 0)) > 0$ at the boundary of the set, thus if $r > 0$ there always exists an $\varepsilon > 0$ such that $f(\varepsilon, y) > f(0, y)$ and (ε, y) is feasible.
- (e) The minimisation problem is convex if the objective function is concave and the feasible set is concave. For $a = 0$ we have $f(x, y)$ strictly convex and the feasible set is also convex.
- (f) We know that negativity constraints may be ignored, hence

$$L(x, y, \lambda) = x^2 + y^2 + x + 2y + \lambda(x^2 + y^2 - r)$$

hence

$$\begin{aligned} 2x + 1 + 2\lambda x &= 0 \\ 2y + 2 + 2\lambda y &= 0 \\ \lambda(x^2 + y^2 - r) &= 0 \\ \lambda &\geq 0 \end{aligned}$$

Candidates may also keep the inequality constraints, this is not a mistake. The Jacobian is $(2x, 2y)$, so the constraint qualification holds at every point except $(0, 0)$ where the value of the function is 0.

Now, if constraint does not bind, then $\lambda = 0$ and $x^* = -1/2, y^* = -1$ and $f(x^*, y^*) = -5/4$ (this also can be computed from the second question by setting $a = 0$). As this is the unconstrained minimum of the function, no further arguments are needed.

If $\lambda > 0$ from the first two FOCs we get: $y = 2x$ and hence $x^* = -\sqrt{r/5}$ $y^* = -\sqrt{r/5}$. This gives $f(x^*, y^*) = \sqrt{r/5}(\sqrt{r/5} - 3)$. Checking condition $\lambda = \frac{1-2\sqrt{r/5}}{\sqrt{r/5}} > 0$ gives $r < 5/4$. Note, that this is precisely the radius of the

circle passing through unconstrained maximum $(-1/2, -1)$. Students should argue why this is the minimum, for example, by checking the bordered Hessian:

$$\begin{vmatrix} 0 & 2x & 2y \\ 2x & 2+2\lambda & 0 \\ 2y & 0 & 2+2\lambda \end{vmatrix} = -8(1+\lambda)(x^2+y^2) < 0$$

5. (a) If the effort is observable, then implementing high effort cost $\bar{u}^2$ for e_0 and $(\bar{u}+c)^2$ for e_1 . Then, $E(x|e_0) = 95$ and $E(x|e_1) = 110$. Implementing high effort is profitable if

$$110 - (\bar{u} + c)^2 \geq \max\{95 - \bar{u}^2, 0\}$$

This implies $c < \min\{\sqrt{110} - \bar{u}, \sqrt{\bar{u} + 15} - \bar{u}\}$, where $\bar{u} > \sqrt{110}$.

- (b) In this case the optimal contract is first best, $w_i = \bar{u}^2$ for each level of catch.
(c) Let v_i , $i = l, m, h$ be the square root of the salary paid in the case the catch is 60, 100, 150 respectively. Then,

$$\begin{aligned} IC : \quad & \frac{1}{6}v_l + \frac{1}{2}v_m + \frac{1}{3}v_h - c \geq \frac{1}{3}v_l + \frac{1}{2}v_m + \frac{1}{6}v_h \\ IR : \quad & \frac{1}{6}v_l + \frac{1}{2}v_m + \frac{1}{3}v_h - c \geq \bar{u} \end{aligned}$$

The principal's problem is to

$$\max 110 - \frac{1}{6}v_l^2 - \frac{1}{2}v_m^2 - \frac{1}{3}v_h^2 \quad \text{s.t. } IC, IR$$

Students may directly solve optimization problem or simplify constraints as

$$\begin{aligned} v_h &= v_l + 6c \\ v_m &= 2(\bar{u} - c) - v_l \end{aligned}$$

and then plug it into the principal's problem:

$$\min \frac{1}{6}v_l^2 + \frac{1}{2}(2(\bar{u} - c) - v_l)^2 - \frac{1}{3}(v_l + 6c)^2$$

By taking F.O.C. we obtain:

$$v_l = \bar{u} - 3c, \quad v_m = \bar{u} + c, \quad v_h = \bar{u} + 3c$$

and corresponding wages are

$$v_l = (\bar{u} - 3c)^2, \quad v_m = (\bar{u} + c)^2, \quad v_h = (\bar{u} + 3c)^2.$$

- (d) In case of observable effort the principal prefers e_1 as $110 - (6+1)^2 = 61 > 95 - 6^2 = 59 > 0$ (this can also be obtained from conditions derived in part a). If the effort is unobservable, then the wages for e_1 are 9, 49, 81, and the expected cost is 53. As $110 - 53 = 57 < 59$, then principal prefers e_0 .

6. (a) The expectation of exponential distribution is n . Students may show it by deriving the density $f_n(u) = \frac{1}{n}e^{-u/n}$ and then getting

$$Eu = \int_0^\infty \frac{1}{n}ue^{-u/n}du = \frac{1}{n} \left(-nue^{-u/n} \Big|_0^\infty + n \int_0^\infty e^{-u/n}du \right) = \frac{1}{n}n^2 = n.$$

- (b) I expect students to derive the formula for the surplus, if this is not done, they should lose about 20% of the marks for this question. The derivation uses the indifference condition:

$$z_n = -s_n + F(z_n)z_n + [1 - F(z_n)]E(u|u \geq z_n) = -s_n + F(z_n)z_n + \int_{z_n}^\infty uf_n(u)du,$$

which gives

$$\int_{z_n}^\infty (u - z_n)f_n(u)du = s_n.$$

Some students may jump directly to this reduced formula. Computing the integral using the same approach as the expectation gives

$$\int_{z_n}^\infty (u - z_n)f_n(u)du = ne^{-z_n/n}.$$

This implies that

$$z_n = n \log \frac{n}{s_n}.$$

- (c) The optimal search rule is standard from the lecture. If the largest reservation value of unopened boxes is below the best prize so far, stop the search. If it is above, then open the box with largest reservation value.
- (d) We get that $z_n = -n \log a$. If $a > 1$, then the reservation value for all boxes is negative (search cost exceeds the expected value!), so Pandora should not search. If $a = 1$, then for all n we have $z_n = 0$ and Pandora is indifferent opening and stopping. So, she either does not open any boxes, or opens one and stops with probability 1. If $a < 1$ then Pandora will be opening boxes starting from the last one and working towards the first one, until the best discovered prize exceeds the highest reservation value of unopened boxes..
- (e) Here I want students to use Armstrong (2017) covered in the lectures, proof is not required (students who derived it should get extra credit). The result says that Pandora will choose the box with the largest virtual value $v_n = \min\{z_n, u_n\}$, where u_n is the actual prize. In this case $z_n = n$, so the box with the largest $\min\{n, \frac{10000}{n}\}$ is box $n = 100$. So, Pandora will stop once she opens boxes 10000, 9999, ..., 100.